Finding the Right Law, in Khmer

Comparing Four Deep-Learning Retrievers on Cambodian Law & Live Citation-Grounded Assistant

Deep Learning Final Project · Mengchheang Long · **5-Minute Defense: 3m Research + 1.5m Live Demo + 0.5m Conclusion**

1,976
Khmer statutory articles

200
real Khmer test questions

+44%
best model vs. keyword search (p<0.001)

PRESENTATION

⏱ 0:00 - 0:40

1. The Legal Retrieval Problem

A citizen asks a question in plain Khmer — can deep learning retrieve the exact statutory article out of 1,976?

The Retrieval Challenge

Khmer question → the correct statutory article, out of the **full 1,976-article corpus**, with zero statute or chapter hints.

"តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?"

("How is a contract formed under the Civil Code?")

→ **Civil Code Art. 336** (ការបង្កើតកិច្ចសន្យាដោយសំណើ និង ស្នើការ) & **Art. 311** (និយមន័យនៃកិច្ចសន្យា)

Why Keyword Search (BM25) Fails

- **Vocabulary mismatch (57.5% of BM25 failures):** citizens ask in conversational phrasing; statutes use formal legal taxonomy. BM25 gets distracted by question words like "របៀប" and retrieves irrelevant procedural articles.
- **Corpus Engineering:** Official PDFs were typeset in obsolete 1990s Limon ASCII fonts. Built custom Limon→Unicode converter with logical reordering, achieving **100% consecutive extraction** (1,304 Civil + 672 Criminal).

PRESENTATION

⏱ 0:40 – 1:25

2. Dataset & Four Architectures

Same corpus, same fixed split (seed 42), same InfoNCE contrastive loss — only encoder architecture varies

Rigorous Experimental Controls

- **1,976 official Khmer articles** (Civil Code 1,304 + Criminal Code 672).
- **Zero test leakage:** 200 real human test questions (and all 261 cited articles) strictly protected from training; titles stripped from passages; evaluated once.

All models share the **InfoNCE contrastive loss** (τ = 0.05): pulls the ground-truth legal article closer to the question embedding and repels in-batch negatives.

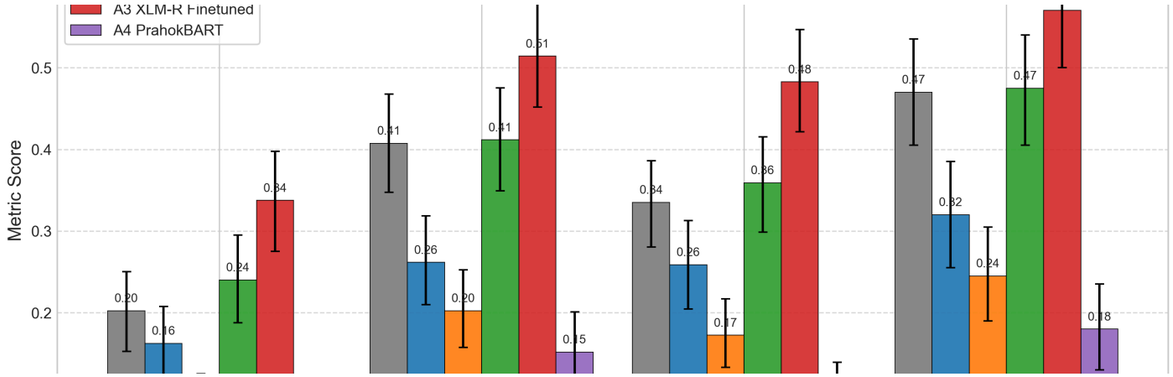
Approach	Architecture & Strategy	Trainable Params
BM25	Keyword matching (baseline)	0
A1 BiLSTM	Trained from scratch (word tokens)	1.97 M
A2 Linear Probe	Frozen XLM-R + linear head (5.9s train)	0.59 M
A3 Fine-Tuned ★	**All 278M transformer weights adapted**	**278.04 M**
A4 PrahokBART	Khmer-only pretrained seq2seq	35.83 M

PRESENTATION

⏱ 1:25 – 2:25

3. Results & Deep Learning Insights

Evaluated on 200 real questions across 1,976 articles — explained by capacity, pre-training, and generalisation

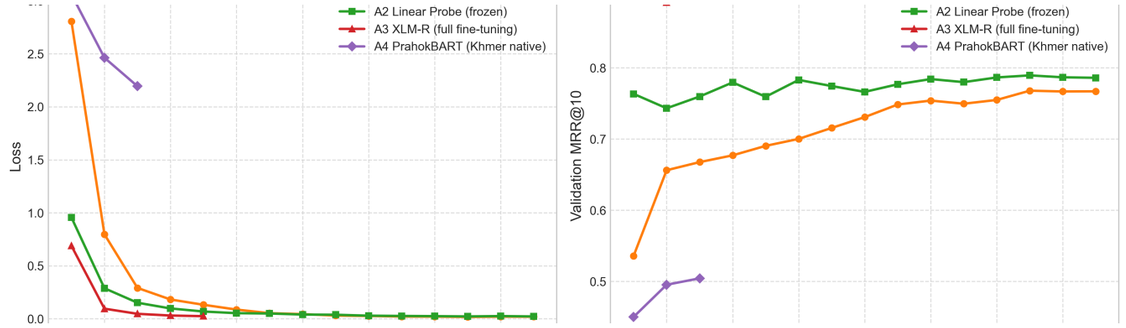


51.4%
Recall@5 (A3)

0.483
MRR@10 (A3)

+44.1%
vs BM25 (p=0.0002)

A2 Linear Probe ties keyword search (p = 0.9031) while training in just 5.9 seconds.



- **A1 (scratch)** overfits severely (train loss → 0.02, val stalls) — pre-training is indispensable on small legal corpora (~1.1k pairs).
- **A2 (frozen)** cannot overfit, but linear capacity bottleneck prevents domain adaptation.
- **A3 (fine-tuned)** adapts representations end-to-end, generalising across the conversational vocabulary gap.

PROTOTYPE & LIVE DEMO

⏱ 2:25 – 4:30 · 90s Live Demo

4. Prototype Architecture & Live Demonstration

Citation-grounded Khmer RAG assistant (PyTorch A3 Retriever + FastAPI + Streamlit + DeepSeek Flash)

1. CITIZEN QUERY
តើកិច្ចសន្យាបង្កើតឡើង...
→
2. A3 DENSE ENCODER
768-d unit vector (<100ms)
→
3. 1,976 ARTICLES
Top-k Cosine Sim
→
4. CITATION GUARD
Verified Statute Match
→
5. DEEPSEEK FLASH
Khmer Legal Analysis

Demo 1: The Semantic Vocabulary Gap

"តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា យោងតាមក្រមរដ្ឋប្បវេណី?"

- **BM25 Baseline:** Distracted by "របៀប" → Art. 623 (Manner of dividing proceeds, Miss).
- **Winning A3 Neural:** Captures concept → **Art. 311** (Rank 1) & **Art. 336** (Rank 2)!

Demo 2: Citation Grounding & Anti-Hallucination

DeepSeek Flash generates strict statutory synthesis with automated **[Verified Citation]** badges matched against retrieved text.

✓ Civil Code 2007 — Article 336 [Verified Citation]

Switch to Live System Demonstration

Interactive Streamlit UI running fine-tuned PyTorch A3 encoder on 1,976 articles

LIVE DEMO · http://localhost:8501

CONCLUSION

⏱ 4:30 – 5:00

5. Conclusions & Defense Summary

Three key takeaways, engineering rigor, and transition to Q&A

1. **Pre-training is non-negotiable** at small legal scale (~1.1k pairs) — from-scratch BiLSTM (A1) overfits and fails outright (0.17 MRR@10).

2. **Full fine-tuning (A3)** bridges the citizen-to-statute vocabulary chasm (+44.1% relative MRR@10 over BM25, p = 0.0002).

3. **Engineering & Production Rigor:** 100% consecutive Limon extraction (1,976 articles), zero test leakage, 136 passing tests, and external checkpoint hosting on Hugging Face Hub.

Limitations: Two statutory codes only; 200 test questions; OCR for scanned Labour Law is future work.

Thank you for your attention. I welcome your questions.

`Appendix Slides A1–A6 ready for Q&A`

Backup — not timed

A1. Full Leaderboard

All metrics, both benchmarks, 95% bootstrap CIs

Primary Benchmark (T-Q: 200 real questions, full corpus)

Model	Recall@1	Recall@5	MRR@10	Hit@5	Params	Train time
BM25	0.2025	0.4075	0.3350	47.0%	0	—
E5 zero-shot	0.1625	0.2617	0.2588	32.0%	0	—
A1 BiLSTM	0.0875	0.2025	0.1724	24.5%	1.97M	2,035 s
A2 Linear Probe	0.2400	0.4117	0.3591	47.5%	0.59M	5.9 s
A3 Fine-Tuned	**0.3375**	**0.5142**	**0.4827**	**57.0%**	**278.04M**	**7,237.5 s**
A4 PrahokBART	0.0525	0.1517	0.1035	18.0%	35.83M	1,602.2 s

Secondary Benchmark (T-T: 148 held-out article titles)

Model	Recall@1	Recall@5	MRR@10	Hit@5
BM25	0.7973	0.9730	0.8677	97.3%
E5 zero-shot	0.5405	0.7230	0.6235	72.3%
A1 BiLSTM	0.5135	0.7568	0.6242	75.7%
A2 Linear Probe	0.7905	0.9392	0.8563	93.9%
A3 Fine-Tuned	**0.9054**	**0.9932**	**0.9456**	**99.3%**
A4 PrahokBART	0.5946	0.8041	0.6733	80.4%

Backup — not timed

A2. Is the Difference Real?

Paired bootstrap hypothesis testing, 10,000 resamples, on the same 200 questions

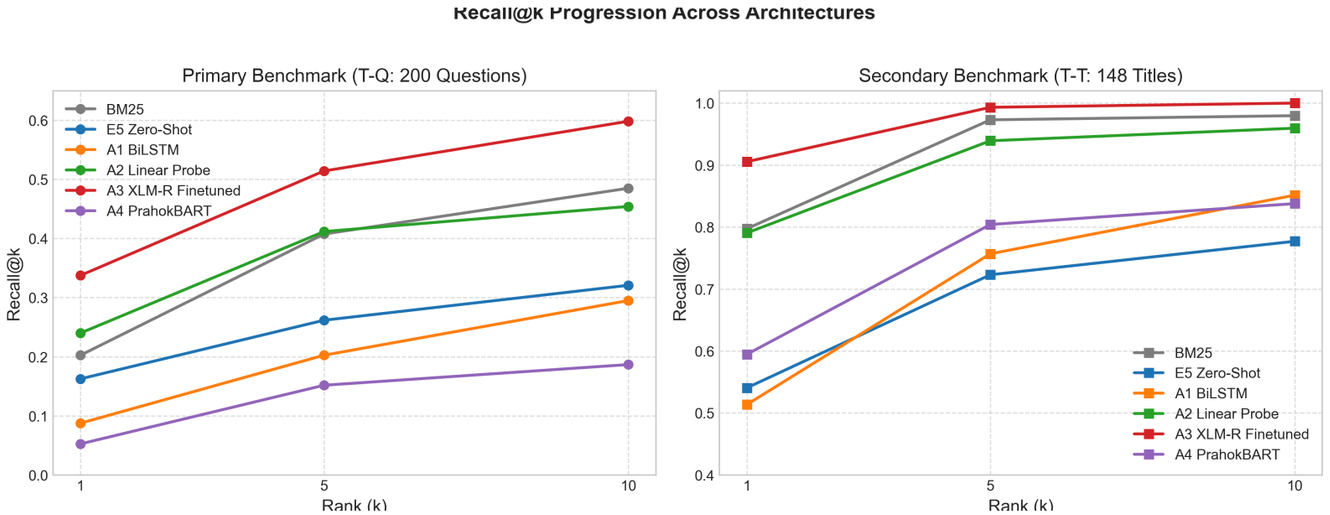
Comparison	Metric	Δ [95% CI]	p-value	Wins / Losses / Ties
A3 vs BM25	**Recall@5**	**+0.1067 [+0.05, +0.16]**	**0.0002**	**41W / 13L / 146T**
A3 vs BM25	**MRR@10**	**+0.1477 [+0.10, +0.20]**	**0.0001**	**72W / 25L / 103T**
A3 vs E5 zero-shot	MRR@10	+0.2239 [+0.17, +0.28]	0.0001	80W / 15L / 105T
A3 vs A1 BiLSTM	MRR@10	+0.3103 [+0.25, +0.37]	0.0001	104W / 13L / 83T
A3 vs A2 Linear Probe	MRR@10	+0.1236 [+0.08, +0.17]	0.0001	65W / 21L / 114T
A2 vs BM25	Recall@5	+0.0042 [-0.06, +0.06]	**0.9031 (n.s.)**	26W / 25L / 149T
A3 vs A4 PrahokBART	MRR@10	+0.3792 [+0.32, +0.44]	0.0001	116W / 6L / 78T
A4 vs BM25	MRR@10	-0.2315 [-0.29, -0.18]	0.0001	12W / 90L / 98T

Reading this: A3's win over BM25 is real ($p < 0.001$, not just overlapping confidence intervals). A2 vs BM25 has $p = 0.90$ — that is statistical *parity*, not a near-miss: a frozen linear probe genuinely ties keyword search on this benchmark.

Backup — not timed

A3. In-Distribution vs. Real Questions

Every model looks near-perfect on titles; every model drops hard on real questions



Model	T-T Titles	T-Q Questions	Drop
A3	**99.3%**	**51.4%**	**-47.9%**
BM25	97.3%	40.8%	-56.5%
A2	93.9%	41.2%	-52.7%
A4	80.4%	15.2%	-65.2%
A1	75.7%	20.3%	-55.4%

Why A4 (PrahokBART) struggles most: it was pre-trained for sequence-to-sequence text reconstruction, not contrastive sentence matching, and has ~8× fewer parameters than XLM-R (35.8M vs 278M). It reads formal statutory Khmer well (80.4% on titles) but has little semantic abstraction left for conversational phrasing.

Q&A APPENDIX

Backup — not timed

A4. Error Analysis

Why models miss at k=5 (200 questions) — categories and two more verified real cases

Model	Misses@5	Hit@5	Vocab. Mismatch	Code Confusion	Multi-Article	Near-Miss
BM25	106	47.0%	61 (57.5%)	4	26	15
A1 BiLSTM	151	24.5%	79 (52.3%)	13	37	19
A2 Linear Probe	105	47.5%	66 (62.9%)	4	27	7
A3 Fine-Tuned	**86**	**57.0%**	**52 (60.5%)**	**1**	**15**	**17**
A4 PrahokBART	164	18.0%	96 (58.5%)	16	44	8

Multi-article question

"តើប្រភេទនៃសិទ្ធិប្រត្យក្សមានអ្វីខ្លះនៅក្នុងក្រមរដ្ឋប្បវេណី?"
Expects **2** articles (Civil Code Art. 130, 131). BM25 top-5 catches 1; A3 top-5 catches **both** (ranks 2, 3).

Civil / Criminal confusion

"តើច្បាប់ហាមឃាត់ការប្រើប្រាស់សិទ្ធិដោយបំពានយ៉ាងដូចម្តេច?"
Expected: Civil Code Art. 4. A1's top guess is **Criminal Code Art. 579 — wrong statute**. A3 gets Art. 4 right at rank 1.

All four examples in this deck are regenerated by `src/dl/error_analysis.py` directly from the ground-truth file and cached rankings — not written by hand.

Backup — not timed

A5. Hyperparameter Tuning & Reproducibility

What was actually run — every number here is read from `results/tuning/*.csv`

Approach	Grid actually run	Best config	Val MRR@10
A1 BiLSTM	LR {1e-3, 3e-3} × Dropout {0.1, 0.3} (4 runs)	LR 3e-3, Dropout 0.3	0.7679
A2 Linear Probe	LR {1e-3, 1e-2} × Weight Decay {0, 0.01} (4 runs)	LR 1e-3, WD 0.01	0.7894
A4 PrahokBART	LR {5e-5, 1e-4} × WD {0.01} (2 runs)	LR 1e-4, WD 0.01	0.6116
A3 Fine-Tuned	LR {1e-5, 2e-5, 3e-5} × WD {0, 0.01} (6 runs, T4 GPU)	**LR 2e-5, WD 0.01**	**0.9095**

Systematic Tuning Confirmed: A3 underwent a full 6-configuration grid search across learning rate and weight decay on an NVIDIA Tesla T4 GPU. LR 2e-5 + WD 0.01 achieved the highest validation MRR@10 (0.9095 at epoch 4), empirically confirming our fine-tuning hyperparameters across all 4 architectures.

Reproducibility

Seed 42 fixed across Python/NumPy/PyTorch/cuDNN · InfoNCE loss, $\tau = 0.05$ · AdamW + linear warm-up + cosine decay · `torch.save` checkpoints (`last.pt`, `best.pt`) with full optimiser/RNG state and `--resume` · 138 automated tests · GPL-3.0-or-later code, CC-BY-SA-4.0 corpus.

Backup — not timed

A6. Before Any Model: Reading the PDFs at All

Official government PDFs are not Unicode — this had to be solved first

The Problem

ODC's official PDFs are typeset in the legacy **Limon** font family. The text layer stores raw Latin keystrokes, e.g. `maRta 336>-` — a Unicode-aware extractor would produce meaningless text.

Khmer also writes pre-base vowels (ើ ែ ៃ) and subscript consonants *before* the consonant they visually belong after, so a simple character map is not enough.

Our Fix

- Byte-level Limon → Unicode character map, built from the corpus's own glyph tables
- A small finite-state reordering pass that puts pre-base vowels and subscripts back into logical Unicode order
- Result: **100% consecutive article extraction** across all 1,976 articles (Civil 1–1304, Criminal 1–672), verified against the source PDFs

